

Regression, pt. 1 (REVISED)

CSU Chico, Math 351

2026-04-20

outline

Method of Least Squares

- Introduction

- Least Squares Equation

- Example

Simple Linear Regression

- Visualization

- parameters

- residuals

- estimation

References

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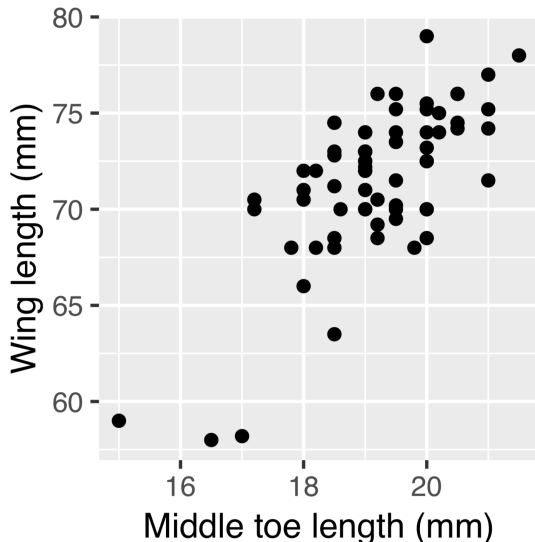
Introduction

Least squares is a method for constructing a predictor of one of the variables from the other by making use of a sample of observed pairs. Many of the fundamental ideas associated with the study of relationships can be nicely illustrated when only two variables are involved.

When we only have two variables involved we say that we have paired data $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. Simple linear regression attempts to model the relationship between the x and y variables with a straight line. Of course, we should look at a graph of the two variables before deciding whether this is an appropriate thing to do.

Scatterplots

Scatterplots are a graphical description of two numerical variables.
Consider Darwin's "finches" data:



Scatterplots

We used some keywords to describe scatterplots.

- ▶ associated or not.
- ▶ direction: positive or negative association.
- ▶ structure: linear or nonlinear.

What kind of association do the two variables on the previous page appear to have?

Finding the Best Line

If a line appears appropriate to describe the data, then we'll need two parameters to estimate for the equation of the line: **slope** and **intercept**.

You **could** arbitrarily draw a line through the data and estimate the slope and y-intercept that way—but how can we determine if it's better or worse at predicting the data than another, similarly arbitrary line?

Finding the Best Line

One method of constructing a straight line fit to the observed values is called the *method of least squares*. For each observation, we can calculate the distance (in y) from the **observed** value to the **predicted** value by the line. This vertical distance is referred to as a **residual**—the error you get when you use the line to predict a value of y .

Thus, a line with small residuals would be better than a line with large residuals. The idea behind least-squares regression is that it finds the line that minimizes the **squared** residuals.

Note that a residual is specifically **observed minus predicted**, so it being positive or negative indicates an under- or over-estimation, respectively.

Least Squares Equation

Let $(x_1, y_1), \dots, (x_n, y_n)$ be a set of n points. The straight line, $y = a + bx$ that minimizes the sum of the squares of the vertical deviations of all the points from the line has the following slope and intercept:

$$b = \frac{n \sum_{i=1}^n x_i y_i - (\sum_{i=1}^n x_i) (\sum_{i=1}^n y_i)}{n (\sum_{i=1}^n x_i^2) - (\sum_{i=1}^n x_i)^2}$$

$$a = \frac{\sum_{i=1}^n y_i - b \sum_{i=1}^n x_i}{n} = \bar{y} - b\bar{x}$$

Example: Air Conditioning (from Larsen & Marx)

An air conditioning manufacturer is monitoring the weight (in ounces) of a component's rough casting and its finished form to see if a linear relationship exists between the two (see the "aircon.csv" data set in the CodiMD notes). Using the Least Squares Equation, identify the slope and intercept of the best straight line approximating the data.

Example: Air Conditioning (from Larsen & Marx)

Some key quantities will need are: $\sum x_i$, $\sum y_i$, $\sum x_i^2$, $\sum y_i^2$, and $\sum x_i y_i$. There are a few ways to obtain these from the CSV data, but one possible approach is to use `colSums()`:

```
n <- 25
sums <- as.numeric(colSums(aircon[c(2:3)]))
sums.sq <- as.numeric(colSums(aircon[c(2,3)]^2))
sums.xy <- as.numeric(colSums(aircon[2]*aircon[3]))
```

This will calculate all of the sums in question, at which point we can find the slope and intercept:

```
(b <- (n*sums.xy - sums[1]*sums[2])/
(n*sums.sq[1] - sums[1]^2) )
(a <- (sums[2]-b*sums[1])/n )
[1] 0.6420981
[1] 0.3077347
```

Example: Air Conditioning (from Larsen & Marx)

The resulting least-squares line is:

$$y = a + bx = 0.3077 + 0.6421x$$

This can be used to predict finished weights based on rough weights. A rough weight (x) of 2.71 oz would imply an estimated finished weight (y) of:

$$y = 0.3077 + 0.6421(2.71) = 2.048\text{oz}$$

We'll discuss the interpretation of the slope and intercept in the next section.

Exercise: Finches Data

Considering the finches data from before that shows how wing length and middle toe length are related, identify the least squares line for the relationship between the two (let x = middle toe length and y = wing length). The data can be found on today's CodiMD notes.

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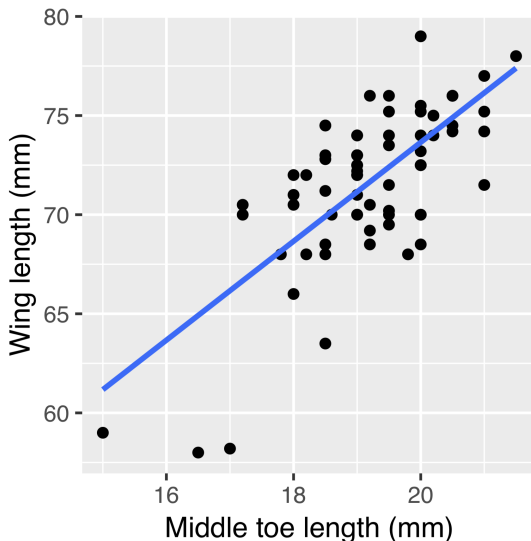
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Simple Linear Regression, introduction

The previous section was from a purely geometric perspective. To get a more statistical approach, we can think of each observed y as the value recorded for a random variable Y so that a distribution of possible y -values is associated with every value of x . In other words, a given x is not expected to produce the exact same y every time—some error is expected.

Simple Linear Regression, plotted

The **least squares line** for Darwin's finch data is plotted below in blue.



Simple Linear Regression, plotted

The coefficients of the **least squares line** for Darwin's finch data are

```
##      (Intercept) middletoelength
##      23.751547      2.494932
```

Simple Linear Regression, idea

Simple linear regression decomposes the response variable Y into three components:

- ▶ the **intercept**
 - ▶ the value Y takes on when X is equal to 0;
 - ▶ above, the length of a wing when the middle toe length is 0
- ▶ the **slope**
 - ▶ on the explanatory variable X
 - ▶ represents the increase in Y for a unit increase in X ;
 - ▶ above, some increase in wing length for every mm increase in the middle toe length
- ▶ **errors/residuals**
 - ▶ some left over bits

Linear Regression, model

Given a response variable Y and an explanatory variable X , the simple linear regression model is

$$Y_i = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} X_i + \underbrace{\epsilon_i}_{\text{errors}}, \quad \epsilon_i \sim N(0, \sigma^2),$$

where $i = 1, \dots, n$.

Simple Linear Regression, parameters

The population parameters β_0 and β_1 are estimated with $\hat{\beta}_0$ and $\hat{\beta}_1$. These estimates within the linear regression equation are written

$$E(Y|X) = \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X.$$

The expected value, or fitted value, of Y is a function of the estimates of the intercept and slope, dependent on some value of X .

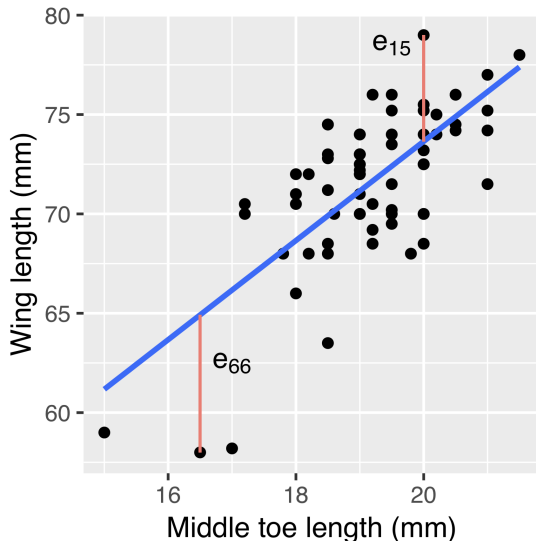
Linear Regression, residuals

Not every observation will fall on the least squares line. The difference between the true observation Y_i and the predicted value of $\hat{Y}_i$ at the X_i , is the i th residual

$$e_i = Y_i - \hat{Y}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$$

Simple Linear Regression, residuals by picture

Some residuals will be positive and some negative.



Simple Linear Regression, best

As seen earlier, our ideal line minimizes the sum of the squared residuals and will have the values of β_0 and β_1 that

- ▶ take $e_i = Y_i - \hat{Y}_i$, for all i ,
- ▶ square each residual, e_i^2 , and
- ▶ minimize $\sum_{i=1}^n e_i^2$.

Simple Linear Regression

Our model implies $Y \sim N(\beta_0 + X\beta_1, \sigma^2)$. Parameter estimates, $\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2$, are found via likelihood.

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references I

Hadley Wickham. *ggplot2: elegant graphics for data analysis*. Springer Science and Business Media, 2009.

David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel. *OpenIntro Statistics*. CreateSpace independent publishing platform, third edition, 2015.

Richard J Larsen, Morris L Marx. *An Introduction to Mathematical Statistics and Its Applications*. Prentice Hall, fifth edition, 2012.

Adapted from notes by Dr. Gray & Dr. Roualdes.